A Kernel-Checked Spectral-to-Tiling Theorem for the Cyclic Group $\mathbb{Z}/180\mathbb{Z}$

Javier Emilio Bazán Sanchez
Facultad de Ciencias, Universidad Nacional Autónoma de México (UNAM)
bazan@ciencias.unam.mx

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Abstract. We give a machine-checked proof of the spectral-to-tiling implication for the cyclic group $\mathbb{Z}/180\mathbb{Z}$. Spectrality is expressed exactly: the mask polynomial of a finite set is required to have the cyclotomic divisors corresponding to all nonzero differences in a proposed spectrum. The conclusion is a genuine translational tiling, encoded as bijectivity of the addition map. The final obstruction occurs at cardinality 30. Its marginal component is reduced from 16,796 literal pairs to 213 profile cells through an ordered-difference histogram identity; 16,574 pairs are refuted by an integer scalar coefficient and the remaining 222 are connected to explicit catalogue witnesses. A separate common-frame argument handles the coupled five-Gram data. These certificates are joined to the global cardinality reduction in Lean 4. The final theorem and its principal intermediate endpoints replay successfully with `leanchecker`; their printed dependencies are only `propext`, `Classical.choice`, and `Quot.sound`. The full source, deterministic generators, manifests, and a 946-module replay plan accompany the article.

1 Introduction

Fuglede's conjecture relates two rigid structures: a set that admits an orthogonal basis of characters and a set that tiles its ambient group by translations. In Euclidean spaces the unrestricted conjecture is false in high dimension, while important finite and one-dimensional cases retain a rich arithmetic structure [4, 7, 3]. Finite cyclic groups are especially useful because Fourier vanishing can be encoded by divisibility of integer mask polynomials by cyclotomic polynomials [1]. This turns analytic orthogonality into an exact, checkable statement about polynomials and roots of unity [5, 6, 9].

This article reports a complete formal proof of one such finite statement for $G = \mathbb{Z}/180\mathbb{Z}$. The modulus

$$180 = 2^2 \cdot 3^2 \cdot 5$$

is small enough to permit an explicit classification, but composite enough to exhibit several interacting prime-power directions. Most cardinalities are removed by structural reductions. The difficult residual case has $|A| = 30$ and requires both a marginal classification and compatibility of a five-row Gram configuration.

The central design choice is to make the finite computation part of the proof, rather than an oracle beside it. Scripts generate ordinary Lean declarations; the Lean kernel checks the generated proofs. Hash manifests fail closed if sources, catalogues, witness locations, or import order change. No floating-point calculation enters the theorem.

Contribution

The formal development establishes the following implication.

Theorem 1 (Formal spectral-to-tiling theorem)**.** *Let $A, L \subseteq \mathbb{Z}/180\mathbb{Z}$. If L is an exact cyclotomic spectrum for A, then there exists $B \subseteq \mathbb{Z}/180\mathbb{Z}$ such that the map*

$$A \times B \longrightarrow \mathbb{Z}/180\mathbb{Z}, \qquad (a, b) \longmapsto a + b$$

is bijective.

In the repository, Theorem 1 is the declaration `Fuglede.z180_exists_tiling_of_spectral_v96`.

2 The exact finite formulation

For $A \subseteq \mathbb{Z}/N\mathbb{Z}$, choose the standard representatives $a \in \{0, \dots, N-1\}$ and define the mask polynomial

$$m_A(X) = \sum_{a \in A} X^a \in \mathbb{Z}[X].$$

For $d \in \mathbb{Z}/N\mathbb{Z}$, let

$$\mathrm{ord}_N(d) = \frac{N}{\gcd(N, d)}.$$

The formal predicate `CyclotomicZero` requires

$$\Phi_{\mathrm{ord}_N(d)}(X) \mid m_A(X) \quad \text{in } \mathbb{Z}[X].$$

This is the exact algebraic certificate used for Fourier vanishing at frequency d.

Definition 2 (Exact cyclotomic spectrum)**.** A finite set $L \subseteq \mathbb{Z}/N\mathbb{Z}$ is a cyclotomic spectrum for A if $A \neq \varnothing$, $|A| = |L|$, and for every two distinct $\ell_1, \ell_2 \in L$,

$$\Phi_{\mathrm{ord}_N(\ell_1 - \ell_2)} \mid m_A.$$

The tiling predicate is deliberately independent of Fourier analysis. For finite sets $A, B \subseteq G$, let

$$f_{A,B} : A \times B \longrightarrow G, \qquad f_{A,B}(a, b) = a + b.$$

Lean defines

$$\mathrm{Tiles}(A,B) \iff f_{A,B} \text{ is bijective.}$$

Thus the conclusion includes both existence and uniqueness of each representation. In particular, it implies $|A||B| = |G|$.

The exact Lean endpoint is short enough to display in full.

```
theorem z180_exists_tiling_of_spectral_v96
    {A L : Finset (ZMod 180)}
    (hSpec : CyclotomicSpectrum 180 A L) :
    exists B : Finset (ZMod 180), Tiles A B := by
  exact z180_exists_tiling_of_spectral_of_k30_catalogue_v81
    z180_k30_exceptional_catalogue_completeness_v96 hSpec
```

The work therefore lies in constructing the exceptional catalogue completeness object consumed on the last line.

3 Proof architecture

Figure 1 shows the dependency structure. The global reduction isolates the cardinality-30 obstruction. That obstruction has a projective/marginal component (V97) and a coupled common-frame component (V95). V81 turns their catalogue into the global theorem, and V96 supplies the two certificates unconditionally.

3.1 The global reduction

The surrounding library converts the spectrum hypothesis into exact cyclotomic zero data, applies cardinality and divisor sieves, and closes the nonexceptional branches. The master conditional theorem then asks only for a closure at cardinality 30. The module `Z180K30CatalogueMasterClosureV81` packages this interface: a complete exceptional catalogue implies the spectral-to-tiling theorem for all cardinalities.

This separation is valuable both mathematically and formally. The general Fourier and tiling lemmas do not depend on the large finite catalogue, while the catalogue proof sees a small and rigid normalized state space.

3.2 Projective normalization and ordered differences

The projective branch works with six-element subsets of $\mathbb{Z}/36\mathbb{Z}$. For a list S, define its ordered-difference list

$$\Delta(S) = \big((x-y) \bmod 36\big)_{(x,y)\in S^2}.$$

Diagonal terms are retained. The scalar Gram coefficient used by the audit can be written as the bilinear sum

$$\Gamma(U,V) = \sum_{a\in\Delta(U)} \sum_{b\in\Delta(V)} \zeta_0(ab), \tag{1}$$

where ζ_0 is the exact integer-valued coefficient extracted from the cyclotomic representation. Lean proves Equation (1) by unfolding the Gram trace. Exchanging the two finite sums and using $ab = ba$ proves

$$\Gamma(U,V) = \Gamma(V,U).$$

This symmetry removes a duplicated orientation from the finite audit.

For a supported divisor $d \in \{3,4,6\}$, every projective-side difference is a multiple of d. The U-profile stores the responses

$$K_{d,U}(t) = \sum_{a\in\Delta(U)} \zeta_0(at), \qquad 0 \le t < 36,$$

with zero outside the multiples of d. The V-profile is the sorted list $\Delta(V)$, which is a canonical representation of its histogram. A proved compression lemma states

$$\Gamma(U,V) = \sum_{b\in\Delta(V)} K_{d,U}(b). \tag{2}$$

Sorting changes no sum because Lean carries an explicit permutation proof. The use of a sorted list rather than an unverified hash is important: profile equality is itself checked in the kernel.

3.3 The finite profile census

The normalized literal universe contains 16,796 distinct pairs. Grouping by the two exact profiles produces the much smaller table in Table 1.

Table 1: Authenticated projective-profile census.

d	U profiles	V profiles	cells	positive pairs
3	5	35	175	42
4	4	7	28	0
6	10	1	10	180
Total	19	43	213	222

Each cell evaluates an integer target coefficient. A value different from 936 refutes every pair in that cell. This eliminates 16,574 pairs. A positive value is not accepted merely because the total count is plausible: each of the remaining 222 literal pairs carries an exact pointer into one of the certified V87 witness shards. Lean checks the shard index, witness index, divisor, normalization, and the literal U, V values. The positive counts $42 + 180$ therefore express a set equality, not a probabilistic or numerical match.

The profile compression was also essential for proof engineering. Expanding the original pairwise Gram computation retained millions of repeated root-of- unity reductions and exceeded practical memory limits. Equations (1)–(2) move the repeated work into a small set of reusable profiles without changing the certified proposition.

3.4 The common-frame branch

The marginal catalogue is insufficient by itself: five row configurations must be compatible with the same column frame. The V95 branch handles this coupled condition.

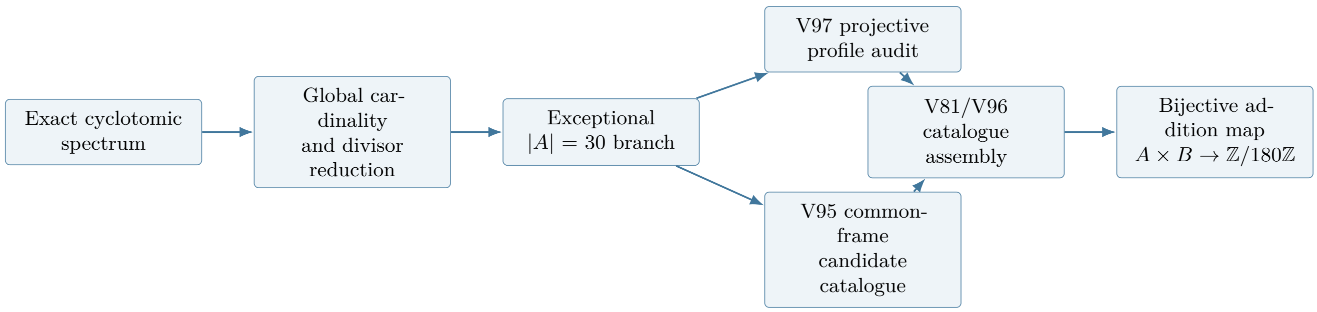

Figure 1: High-level structure of the checked proof. V97 and V95 are independent finite certificates joined only at catalogue assembly.

First, an orbit-frame extractor writes the normalized column as one affine image with a unit and a translation. The same unit is then applied to every row. Covariance lemmas transport the five Gram coordinates into this common frame. Second, a trace classifier identifies the only compatible row orbit. Finally, a bounded affine table supplies a valid candidate with the same exact star signature. Permutation invariance transports the candidate from the representative to the original row.

The result is a function of type `Z180K30ExceptionalFramedStarCatalogueV83`; it contains both the five candidates and the proof that their exact coordinates realize the required star. The finite affine leaves, their aggregate, and the final V95 module are all replayed separately.

3.5 Catalogue assembly

The V97 normalization gives marginal coverage. The V95 framed-star object gives the coupled witness. V83 combines them into complete exceptional catalogue data; V81 converts that data into the global cardinality-30 closure and then invokes the previously completed cardinality reduction. V96 is the final three-import wrapper joining these components.

4 Formal verification and trust

The development uses Lean 4.31.0 and a pinned mathlib revision. Lean and mathlib provide a small trusted kernel and a large reusable mathematical library [2, 8]. The release adopts the following policy.

Exact arithmetic. Mask polynomials have integer coefficients. Finite Gram computations reduce to integer lists and exact cyclotomic coefficients. There is no floating- point tolerance and no numerical root approximation.

Untrusted generation, trusted checking. Python and PowerShell generate literal definitions, theorem statements, and replay schedules. They are convenience tools, not part of the logical trust base. Every generated declaration is elaborated by Lean. Structural roots are replayed again by `leanchecker`.

No proof escape hatches. The authenticated V97 DAG contains no `sorry`, project-defined `axiom`, `native_decide`, or `unsafe`. Ordinary decisions are deliberately kept small; larger statements are split into bounded leaves and composed with proved list and permutation lemmas.

Fail-closed provenance. The master manifest authenticates 946 modules in a unique topological order, including an external algebraic closure of 757 modules. It pins child manifests, generators, catalogue inputs, the order of 66 witness shards, and the 16,796/213/222 census. Source drift causes authentication to stop before Lean is launched.

Axiom report. For both catalogue completeness and the final theorem, Lean prints exactly

[`propext`, `Classical.choice`, `Quot.sound`].

These are the standard classical and quotient principles used by mathlib; no problem-specific axiom appears.

5 Reproducing the artifact

The repository contains the complete Lean project, pinned toolchain, article, certificate inputs, deterministic generators, and manifests. The shortest end-to-end commands are:

```
cd fuglede_lean
lake update
lake build Fuglede.Z180K30UnconditionalCatalogueClosureV96
lake env leanchecker -v Fuglede.
      Z180K30UnconditionalCatalogueClosureV96
cd ..
python scripts/generate_z180_k30_projective_profile_audit_v97.py
```

The last command authenticates the V97 source DAG and manifest. A serialized PowerShell runner is supplied for a module-by-module replay under a fixed memory cap. The public repository records source and checked-object hashes for the V97, V81, and V96 endpoints.

6 Scope and interpretation

Theorem 1 is intentionally precise.

- It proves the spectral-to-tiling direction for $\mathbb{Z}/180\mathbb{Z}$.
- Spectrality is the exact cyclotomic predicate displayed in Section 2; the proof does not assume approximate Fourier vanishing.
- The conclusion is a translational tiling with unique representations.
- The theorem does not assert the converse, a result for arbitrary moduli, or a general theorem for all finite abelian groups.
- The article explains the reduction and certificate architecture; the thousands of finite leaves are best inspected and replayed as code.

This distinction is more than cautious wording. Formal verification certifies that the encoded theorem follows from its definitions; it does not replace the mathematical task of stating those definitions faithfully. For that reason the exact Lean predicates and the public endpoint are included explicitly.

7 Conclusion

The spectral-to-tiling implication on $\mathbb{Z}/180\mathbb{Z}$ can be proved with a small conceptual core and a large but structured finite certificate. The difficult cardinality-30 branch becomes manageable after separating marginal profile data from coupled common-frame data. Ordered-difference profiles compress the marginal audit to 213 exact cells, while explicit witness pointers prevent the positive cases from being accepted by count alone. The common-frame branch then supplies the compatibility missing from a marginal classification.

The resulting V96 declaration is short, but its inputs are fully exposed: exact definitions, algebraic identities, finite tables, generators, hashes, and replay commands are all part of the release. This makes the result both a theorem about a concrete cyclic group and a reproducible case study in turning large finite harmonic-analysis arguments into kernel-checked mathematics.

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